Midterm - Reflective Journal

John Grubbs

I liked this assignment because I got to do all the data processing steps we have talked about so far in this course. Before a model can trained this assignment went over the complete set of steps needed to get the data ready

I was able to spend time examining the importance of data distribution. I also learned how to handle invalid negative billing amounts by changing them to missing values and replacing them with the median from the training data.

The assignment also helped me understand the difference between label encoding and one-hot encoding. I learned that identifying columns, such as Name, Doctor, Hospital, and admission and discharge dates, may not be useful for making predictions. My main takeaway is that the quality of the data-cleaning process can affect how much a machine learning model can be trusted.

Keira Burdine

This project helped me understand the importance of data preprocessing and how it prepares a dataset for machine learning. We chose the healthcare dataset from Kaggle, which contains 55,500 rows and 15 columns, providing enough information to explore different patient characteristics and test results. Using Python in Jupyter Notebook, I performed data exploration, removed duplicate records, handled negative billing amounts, and checked for missing values. I also practiced feature engineering by creating age groups, encoding categorical variables, and scaling numerical features. Using Matplotlib and Seaborn, I visualized the data before and after preprocessing to better understand its distributions. One important lesson I learned was that preprocessing requires careful decision-making because changing data incorrectly can affect the accuracy and reliability of a machine learning model. I also learned that not every dataset needs class balancing, since my three test-result categories were already approximately balanced.

Overall, this project improved my understanding of how to prepare real-world data for machine learning and strengthened my skills in Python, pandas, scikit-learn and data visualization. I learned the importance of keeping the original dataset unchanged and performing all modifications through code. I also realized that splitting the dataset into training and testing sets before preprocessing helps prevent data leakage and supports a more reliable evaluation. In the future, I can use this healthcare dataset to develop a multiclass classification model that predicts whether a test result is abnormal, inconclusive, or normal. I plan to compare models such as Logistic Regression and

Decision Tree using accuracy, macro F1-score, and confusion matrices to evaluate their performance. This experience taught me that successful machine learning involves more than building a model; it also requires cleaning data carefully, selecting appropriate features, and evaluating results responsibly.

Luis Adrian

This midterm project was a good challenge for this semester; this project helped me understand why data preparation an important step is before training a machine learning model. Working with the healthcare dataset allowed me to practice checking data types, missing values, duplicates, and class distributions. I also learned why the dataset must be divided before exploration and preprocessing. Using only the training data for these steps helps protect the testing data for a more reliable final evaluation.

One important lesson was that "preprocessing" decisions should depend on the data rather than applying every technique automatically. The training classes were already approximately balanced, so resampling was unnecessary. The Yeo-Johnson transformation changed the billing distribution but did not improve its symmetry, so the corrected values were restored. Negative billing amounts were treated as missing and replaced with the training median under an explicit project assumption that billing amounts should be non-negative.